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Research Note

The Effect of Return Policy Leniency on Consumer Purchase and Return Decisions: A Meta-analytic Review

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Abstract

While most retail stores offer return policies, some offer more lenient return policies than others. The inherent belief is that lenient return policies are more likely to lead to purchases than to encourage returns. Examining prior research we find that return policy leniency has been characterized in terms of five different dimensions: time, money, effort, scope, and exchange. We conduct a meta-analysis of 21 papers examining the effect of leniency on purchase and return decisions, and demonstrate that overall, leniency increases purchase more than return. Further, we show the return policy factors that influence purchase (money and effort leniency increase purchase) are different from the return policy factors that influence returns (scope leniency increases returns while time and exchange leniency reduces returns).
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Return policies are a consumer risk reliever often used by retailers (e.g., Greatorex and Mitchell 1994) to increase consumer demand. However higher demand also likely leads to higher product returns. Product returns in 2014 totaled about $280 million across all U.S. retailers—a figure that exceeds the annual sales revenue of all but the top ranked retailer.[3] Returns can be prohibitively expensive for retailers due to the high processing costs and low salvage values associated with returned merchandise. In fact, a recent research anecdote suggests that retailers with product return rates in excess of 20 percent often generate zero operating profit.[4]

Despite the cost and prevalence of returns, most retailers offer a return policy hoping that the positive effect on demand will more than offset the negative effect on returns. In a survey of 133 stores in California conducted by Davis, Hagerty, and Gerstner (1998), almost all of the stores had a return policy irrespective of whether they were department stores, specialty stores, or single outlet stores. However, the policies differed in that some stores offered more lenient return policies compared with others. In this regard, the first question we seek to address in the current research is: *do lenient return policies increase product purchase more than product returns?*

Extant scholarly research offers no clear-cut answers. In fact, research exploring the effect of lenient return policies on product purchase and subsequent returns is inconclusive. To illustrate, while Petersen and Kumar (2010) find that lenient return policies lead to both higher purchase and higher returns, Wood (2001) and Wang (2009) find that lenient return policies increase purchase without increasing returns. To provide a more definitive answer to this question, in the current research, we conduct a meta-analysis by synthesizing data from original studies that examine how return policy leniency affects two important downstream outcomes: (1) purchase proclivity (i.e., purchase attitudes

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[4] https://www.theretailequation.com/Retailers/images/public/pdfs/white_papers/wp_TRE4013_WhitePaper_ReturnFraud101_Feb2013.pdf.

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or behaviors) and (2) return proclivity (i.e., consumer intentions or decisions to keep or return products). We show, across studies and on an overall basis, that while return leniency increases both purchase and return, the effect on purchase proclivity is larger than the effect on return proclivity.

In addition to the basic question of whether return policies increase purchase more than returns, a second question we examine is: what type of *return policies have a greater impact on purchase, and what type have a greater impact on returns?* Extant research offers little guidance on this topic. Based on a research synthesis of prior literature, we propose that return policies can be classified as being lenient or restrictive along five dimensions: time leniency (e.g., 60 day vs. 30 day return policy), monetary leniency (e.g., offering 100 percent money back vs. 80 percent money back), effort leniency (e.g., no forms required vs. forms required), scope leniency (e.g., accepting returns on sale items vs. not), and exchange leniency (e.g., cash back vs. store credit). We code the effects of return policy leniency reported in the studies included in our meta-analysis as varying along these five return policy leniency dimensions.

To account for the heterogeneity among papers included in the meta-analysis, we include various substantive moderators (durability of the product, online focus of the retailer, and type of assortment carried) and various methodological moderators (type of journal, type of study, study location, and type of respondent). Using an HLM model that accounts for multiple effects included in each study, we find a distinct set of return policy parameters affect purchase compared with the parameters that affect returns. Specifically, money leniency and effort leniency increase purchase to a greater extent than do the other return policy factors. On the other hand, leniency on time and exchange reduces returns more than other return policy factors, while leniency on scope increases returns. Because the leniency factors have a differential impact on purchase and returns, retailers should consider creating return policies that vary along multiple dimensions.

In the sections that follow, we provide a review of the relevant literature on return policies and their effect on purchase and return proclivities. After delineating the meta-analytic procedure employed and presenting the results, we conclude with a discussion of how our findings provide meaningful contributions to both academicians and retail managers.

Conceptual Foundations

Does Leniency in Return Policies Increase Purchase More Than Returns?

Because most researchers have limited their examinations of return policy leniency and its effect on product purchase or product returns to only one or two return policy factors, prior research does not sufficiently answer this question. For example, a study conducted by Wood (2001) varied only the amount of the monetary refund, whereas Petersen and Kumar (2010) varied only the scope of products, Wang (2009) varied only the amount of time and guarantee, and Janakiraman and Ordóñez (2012) varied only the amount of time and effort. Thus, we know little about the overall effect of return policies on purchase and return.

On the one hand, there is evidence that varying some return policy factors increases purchase without increasing returns. For example, Wood (2001) offered participants the opportunity to order a pen or choose a gift certificate under conditions of either a lenient or a restrictive return policy, which was varied by amount of the monetary refund (i.e., full refund amount for the lenient condition and partial refund amount for the restrictive condition). Upon receipt of the pen and after product trial, participants were asked to decide whether to keep the pen or exchange it for a gift certificate. The lenient return policy was found to increase purchase, but did not result in higher return rates compared with the restrictive return policy. Wang (2009) found similar effects, in that lenient return policies significantly increased initial purchasing tendency but did not increase return rate.

Some previous research shows leniency on other return policy factors likely increases purchase as well as returns. For example, in a controlled field experiment, Petersen and Kumar (2010) varied a company's product return policy by either allowing for non-defective products to be returned (lenient return policy) or only allowing defective products to be returned (restrictive return policy). The average dollar amount of returned products was higher under the lenient return policy ($67.90) than under the strict return policy ($20.50), suggesting a positive effect of lenient return policy on return behavior, with higher levels of return leniency leading to higher return proclivity. There is support for such a notion at the aggregate level as well. Bonifield, Cole, and Schultz (2010) find a positive relationship between number of purchases and number of product returns, suggesting that increasing product purchases leads to greater product returns.

To provide guidance on this topic, we conduct a meta-analysis to examine the overall impacts of return policy leniency, as well as the individual influences of leniency factors that comprise return policies. We classify all measures of purchase intentions and purchase behavior as *purchase proclivity*. Similarly, we group measures of both intentions to return and actual returns as *return proclivity*. A summary of the theories that are relevant to the examination of the effect of return policy leniency on purchase and returns follows.

Mechanisms Driving Effects of Leniency on Purchase

In examining the effect of return policy factors on product purchase, three theoretical mechanisms have been suggested in extant research. First, signaling theory has been put forth to explain how return policies act as positive quality signals by the retailer (e.g., Wood 2001). Second, consumer risk theory has been applied to suggest that return policies should reduce the financial and product risk that consumer's feel prior to product purchase (e.g., Van den Poel and Leunis 1999). Third, construal level theory has been used as a basis for positing that individuals faced with lenient return policies are likely to focus on the benefits of purchase rather than the costs of purchase (e.g., Janakiraman and Ordóñez 2012).

A common thread across the three theories is they all postulate that lenient return policies should positively affect product purchase. However, the theories vary in relevance depending on the return policy factor that is examined. For example, researchers examining monetary leniency or money back guarantees (e.g., Suwelack, Hogreve, and Hoyer 2011; Wood 2001) have employed signaling theory and consumer risk theory, while those examining time deadlines have relied on construal level theory (e.g., Janakiraman and Ordóñez 2012). Intuition suggests that a unit reduction in consumer risk is unlikely to have exactly the same impact on purchase as a unit increase in consumer perceived quality. Thus, we propose that while all return policy factors are likely to have a positive effect on purchase, the degree of their influence will depend on the specific return policy factor that is varied.

Mechanisms Driving Effects of Leniency on Returns

Similar to the empirical evidence reported earlier, the theoretical mechanisms for product returns are mixed in their predictions of the effect of the different return policy factors. Some theories predict lower returns with leniency. For instance, Wood (2001) finds that lenient return policies enhance quality perceptions of the product not only during the pre-purchase stage, but also during the post-purchase stage. This suggests lenient return policies that increase perceived quality should reduce returns. This higher value placed by the consumer on the product increases with longer duration of ownership (i.e., an endowment effect), with longer return deadlines increasing perceived product endowment more than shorter return deadlines (Janakiraman and Ordóñez 2012; Wang 2009; Wood 2001). This suggests lenient return policies that increase endowment should reduce return rates.

However, other theories suggest increased leniency is likely to increase returns. For example, if individuals are sensitive to transaction costs post-purchase (Davis, Hagerty, and Gerstner 1998), then lower transaction costs offered by lenient return policies should lead to higher return rates by making it easier to return a product. Taken together, the various theories suggest that leniency on return policy factors will differ not only by *degree*, but also in terms of the *valence* of effects on product returns.

Typology of Return Policy Factors

Previous researchers have attempted to classify the various elements of return policy leniency, but little consensus exists regarding these factors. Suwelack and Krafft (2012) note that return policies vary significantly depending on their terms. For example, in a survey of 133 stores in California, Davis, Hagerty, and Gerstner (1998) classify return policies in terms of five restriction factors: (1) whether a store provides exchanges or cash refunds; (2) whether a receipt is required or not; (3) whether original packaging is required or not; (4) whether visible signs of use is allowed or not; and (5) the time limit for returns. Alternatively, Heiman, McWilliams, and Zilberman (2001) characterize restrictions in terms of four factors: (1) the length of time allowed for the return (typically between 30 and 60 days); (2) the costs associated with the product return (totally, partially, or not assumed by the retailer); (3) the terms of the policy (e.g., monetary refund vs. a replacement or an exchange); and (4) additional restrictions (e.g., the obligation to have the product returned in its original package). Taking both of these categorization schemes into consideration, we classify return policy leniency as varying along five dimensions:

1. *Time leniency*. Retailers commonly specify deadlines in their return policies (a 30-day policy, a 90-day policy, etc.). Return policies that provide a longer length of time in which to return products are regarded as more lenient.
2. *Monetary leniency*. Lenient return policies allow for a refund of the full monetary amount paid for the product, while strict policies allow for only a portion of the purchase price to be refunded, usually by imposing a "restocking fee" or a non-refundable "shipping and handling fee." Policies that do not impose monetary restrictions are regarded as more lenient.
3. *Effort leniency*. Consumer effort required to execute returns varies, with some retailers creating "hassles" for customers returning products (e.g., requiring the original receipt, tags, or product packaging be retained). Return policies requiring less effort on the part of the consumer are considered more lenient.
4. *Scope leniency*. Stores limit items they consider "return-worthy." For example, products purchased on sale may not be eligible for return. Return policies with greater scope of "return-worthy" items are considered more lenient.
5. *Exchange leniency*. While some retailers offer cash refunds, others offer store credit or product exchange for the returned item. Return policies that allow cash refunds are considered more lenient.

Fig. 1 visually represents the proposed linkage between the return policy leniency factors and the behavioral outcomes of purchase and return.

The systematic coding of the return policy leniency factors varied in each study included in this meta-analysis allows us to address the questions of which leniency factors exert a greater impact on purchase and which have a higher impact on returns.

Substantive Moderators

The extent to which any return policy might affect purchase and returns likely depends not only on the combination of return policy factors employed, but also on the type of retail store that offers it. For example, Van den Poel and Leunis (1999) examined the effect of type of retailer (mail order/internet/specialty store/supermarkets) and type of risk reliever (money back guarantee/price reduction/known brand) as influencers of product likeability. They find a significant interaction effect for store type and risk reliever, suggesting the type of retailer likely influences the effect of various risk relievers. In the current research, we investigate the following substantive moderators (see Fig. 1) relating to suppliers and retailers that are likely to moderate the impact of return policy leniency on purchase and return proclivities:

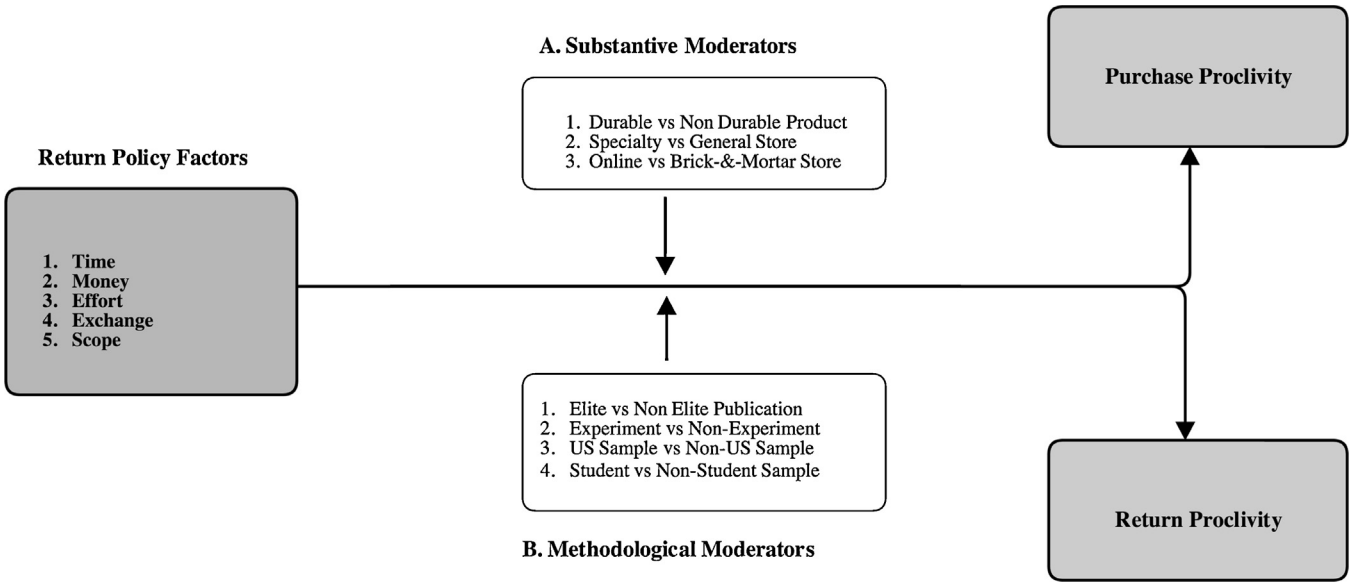


Fig. 1. Meta-analysis model.

1. *Assortment*. The breadth of the assortment carried by retailers varies considerably. In this research, we term retailers carrying a narrow assortment of product categories "specialty stores" and those carrying a broad assortment of product categories "general stores."
2. *Distribution*. Another potential source of variance is the retailer's primary method of distribution. Retailers selling merchandise predominantly through brick-and-mortar stores are classified as "offline focus," while those distributing products through e-commerce web sites are classified as "online focus."
3. *Durability*. Retailers selling products that last a relatively longer time before the consumer repurchases or replaces the product are classified as "durable," while those selling products that are immediately consumed or last a short period of time are categorized as "consumable."

Methodological Moderators

Several methodological variables, which we term methodological moderators, also possess the potential for explaining some variation in leniency-purchase and leniency-return correlations. Studies employing student versus general population samples and U.S. versus non-U.S. samples may vary in purchase and return outcomes. Further, the journal quality (elite vs. non-elite) and the research design (experimental study vs. field study) might also affect purchase and return propensities.

Meta-analysis

Database Development

To build our database, we used multiple approaches to identify the population of relevant research for inclusion in the meta-analysis. First, we conducted a manual search of leading journals (*Decision Sciences, Electronic Commerce Research, European Journal of Social Psychology, International Journal of Retail & Distribution Management, Journal of Applied Psychology, Journal of Applied Social Psychology, Journal of Behavioral Decision Making, Journal of Business Research, Journal of Consumer Marketing, Journal of Consumer Psychology, Journal of Consumer Research, Journal of Economic Psychology, Journal of Experimental Social Psychology, Journal of Fashion Marketing and Management, Journal of Marketing, Journal of Marketing Management, Journal of Marketing Research, Journal of Operations Management, Journal of Product & Brand Management, Journal of Retailing, Journal of Retailing and Consumer Services, Journal of Service Research, Marketing Letters, and Psychology and Marketing*) in which articles investigating the influence of return policies were most likely to appear.

We then searched electronic databases (including EBSCOhost Business Source Complete, Emerald, Google Scholar and Science Direct) with a keyword search, including terms such as "money back guarantees" (e.g., Suwelack, Hogreve, and Hoyer 2011), "lenient return policies" (e.g., Janakiraman and Ordóñez 2012; Wood 2001), "restrictive return policies" (e.g., Bahn and Boyd 2014), "service failures" (e.g., Mollenkopf et al. 2007), and "reverse logistic chains" (e.g., Rogić, Bajor, and Rožić 2010). We also extended our search to unpublished work included in the conference proceedings of the Society for Consumer Psychology and the Association for Consumer Research.

Next, in an effort to prevent a "publication bias," which can reduce the measurement variability in meta-analysis (e.g., Andrews and Franke 1991), we extended our search to unpublished works by manually searching proceedings of additional conferences and contacting authors who have previously published in the return policy domain. Finally, employing an ancestry approach, we examined the references of key conceptual articles and studies identified through the above efforts.

Altogether, these efforts yielded 46 papers with empirical data related to return policies. Based on initial screening of

articles, a study was deemed eligible for inclusion in the meta-analysis if the following information was reported: (1) the Pearson correlation between return policy leniency/strictness and purchase and/or return behaviors or intentions (or enough statistical information was provided to compute the correlation); and, (2) the sample size (Janiszewski, Noel, and Sawyer 2003). We coded 159 effect sizes from 21 papers that met the inclusion criteria.

To detect outliers in the meta-analytic dataset, we utilized the sample-adjusted meta-analytic deviancy (SAMD) statistic recommended by Huffcutt and Arthur (1995), which takes into account both the effect size and study sample size when identifying outliers. The SAMD outlier analysis led us to remove five observations identified as problematic due to extremely large variance and/or calculated correlations near 1.0. As a result, our final dataset consists of 154 observations, of which 75 related to purchase proclivity and 79 related to return proclivity. The total sample size for the studies included in our database is 11,662; however, because our dataset includes multiple effects from various studies, the total sample size for our meta-analytic database is 51,653 (Table 1).

Coding of Return Policy Factors and Moderator Variables

For each effect included in the meta-analysis, three expert raters independently coded the potential moderators—five return policy factors, three substantive moderators, and four methodological moderators—based on information about the study reported in the corresponding paper. A binary coding scheme was applied to each of the variables, with each rater independently rating each effect for the potential moderators. For example, each rater coded the leniency factor variables by inputting a "1" if the return policy factor was manipulated or measured in the study and "0" if it was not manipulated or measured in the study. The specific way each return policy variable was classified as a "0" or a "1" is detailed in Table 2.

The expert raters also coded the type of focal products examined in the study, the type of focal retail store mentioned in the study, and the type of research design on which the study was based. Table 2 details the binary coding scheme used to code the various substantive and methodological moderator variables. There was almost 100 percent agreement among the judges on return-policy factors, and 89 percent agreement on the moderator variables. The coding was compared across the raters and discrepancies were resolved by choosing the coding structure that was chosen by a majority of the coders (i.e., two out of the three coders).

Analytic Procedures

To analyze the impact of return policy leniency on consumers' purchase and return proclivities we conducted both univariate analyses—following procedures described by Borenstein et al. (2009)—and a multivariate approach employing a hierarchical linear model as suggested by Bijmolt and Pieters (2001).

The effect sizes coded for the analyses were zero-order product-moment correlation coefficients, which are easy to interpret and to compare across studies (e.g., Brown and Stayman 1992). We calculated and analyzed the estimated true correlation (r_T) between the return policy leniency factors and key outcomes. In order to calculate the mean r_T, each correlation for a given study was weighted by its sampling variance (which is a function of the sample size for the observation) and then averaged across all studies. Next, the total heterogeneity of the sample (a weighted sum of squares statistic comparable to a total sum of squares in an ANOVA) is calculated and tested against a chi-square (χ^2), distribution to test whether variance among the effect sizes is greater than expected by sampling error (Rosenberg, Adams, and Gurevitch 2000).

A significant chi-square test indicates that inconsistent findings across the observed correlations with respect to product returns cannot be fully explained by statistical artifacts and suggests the presence of other factors (i.e., moderators) in the exploration of heterogeneity (Hunter and Schmidt 2004). The 95 percent bootstrapped confidence interval (CI_{BS}) for each correlation is also calculated in an effort to examine the significance of the mean-corrected correlations. Because meta-analytic data often violate the distributional assumptions of parametric tests, the use of bootstrapped confidence intervals, which are based on a non-parametric distribution, is appropriate and provides a more powerful estimate than traditional confidence intervals (Rosenberg, Adams, and Gurevitch 1997). Finally, the fail-safe sample size (N_{FS}) using Rosenthal's (1979) method was calculated to assess the possibility of publication bias or a "file-drawer" problem. This information estimates the number of unpublished studies with an effect size of zero that would have to exist to render the observed effects insignificant at the alpha equal to a .05 level (Janiszewski, Noel, and Sawyer 2003), with a larger N_{FS} value conveying greater confidence in the results.

Multivariate Analysis using a Hierarchical Linear Model (HLM). Bijmolt and Pieters (2001) suggest the nested nature of meta-analytic data imposes a correlation structure that is best accommodated by specifying a hierarchical linear model (HLM). To jointly assess the influence of the proscribed variables on purchase (return) proclivity we specify a model in which the coded effect sizes are a linear function of the return policy factors and the substantive and methodological moderators. The software package HLM 7 was used to estimate our models of purchase (return) proclivity on leniency factors, retailer factors and manuscript control variables and is formally described in Eqs. (1) and (2).

The variables included in our study of the impact of return policy leniency on purchase and return proclivities vary at the effect-size level (level 1) and the independent sample level (level 2). Due to the limited number of observations in our meta-analytic dataset, we test only for main effects in the HLM.

1. Level 1—Effect Size Model

   The level 1 model explains the variation of effect size i (where $i = 1 \ldots n_j$) around the intercept for study j ($j = 1 \ldots J$). At level 1, the return policy leniency dimension under investigation is captured by p categorical variables where $p = 5$ (*Time*, *Money*, *Effort*, *Scope*, *Exchange*). When the variables are mean centered, π_{0j} represents the mean effect size for

Table 1
Articles used in meta-analysis of return policy effects.

Authors	Year	# of studies	# of observations	Research design	Dependent variable	Product category	Leniency factors examined
Bahn and Boyd	2014	2	170	experiment	intentions	durables	time
Bonifield, Cole, and Schultz	2010	2	541	content analysis; experiment	intentions	consumables	time, money, effort, exchange
Bower and Maxham	2012	2	1,630	longitudinal event field study	behaviors	consumables	money
d'Astous and Guèvremont	2008	1	164	experiment	intentions	durables	time and exchange
Derbaix	1983	1	200	cross-sectional study	intentions	durables	money
Griffis et al.	2012	1	445	experiment	behaviors	durables	effort
Huppertz	2007	1	338	experiment	intentions	durables	effort and scope
Janakiraman and Ordonez	2012	3	430	experiment	behaviors	consumables	time and effort
Javadi et al.	2012	1	107	cross-sectional study	intentions	consumables	time, money, effort, scope, and exchange
Kang and Johnson	2009	1	246	cross-sectional study	behaviors	durables	time, money, effort, scope, and exchange
Kim and Wansink	2012	2	367	experiment	intentions	durables and consumables	time, money, effort, scope, and exchange
Lantz and Hjort	2013	1	4,000	field experiment	behaviors	consumables	money
Maity and Arnold	2013	1	193	cross-sectional study	intentions	durables	time, money, effort, scope, and exchange
Pei, Paswan, and Yan	2014	1	300	cross-sectional study	intentions	consumables	time, money, effort, scope, and exchange
Posselt, Gerstner, and Radic	2008	2	952	content analysis	behaviors	durables	time and money
Powers and Jack	2013	1	308	cross-sectional study	behaviors	consumables	time, money, effort, scope, and exchange
Shao, Chang, and Zhang	2014	1	182	experiment	intentions	durables	money
Suwelack, Hogreve, and Hoyer	2011	2	600	experiment	intentions	consumables	time, money, effort
Van den Poel and Leunis	1999	1	93	experiment	intentions	durables	money
Wang	2009	1	137	experiment	behaviors	consumables	time and money
Wood	2001	3	259	experiment	intentions and behaviors	durables and consumables	time, money, effort, scope, and exchange
Total (21 papers)		31	11,662				

Note: There are 75 effect sizes for purchases drawn from 18 studies, and 79 effect sizes for returns drawn from 17 studies.

Table 2
Meta-analysis coding scheme.

Factor	Variable	Binary coding scheme	
Return Policy	Time	If time deadline varied in study = 1	If time deadline not varied in study = 0
Return Policy	Money	If money back varied in study = 1	If money back not varied in study = 0
Return Policy	Effort	If effort for returns varied in study = 1	If effort for returns not varied in study = 0
Return Policy	Scope	If discounted products accepted for returns = 1	If discounted products not accepted = 0
Return Policy	Exchange	If money is offered on returns in study = 1	If credit or product exchange offered on returns in study = 0
Substantive Moderator	Type of Product	If focal product was durable product = 1	If focal product was not durable product = 0
Substantive Moderator	Assortment Focus	If retail store described in study specialty store = 1	If retail store described in study not specialty store = 0
Substantive Moderator	Channel Focus	If retail store described in study was online store = 1	If retail store described in study was not online store = 0
Methodological Moderator	Journal Quality	If paper published in journal as described by UT Dallas list as elite = 1	If paper published in journal as described by UT Dallas list as not elite = 0
Methodological Moderator	Research Design	If study involved experimental manipulation = 1	If study did not involve experimental manipulation = 0
Methodological Moderator	Study Location	If study was conducted in the United States = 1	If study was not conducted in the United States = 0
Methodological Moderator	Participant Details	If study was among student participants = 1	If study was not among student participants = 0

Table 3
Overall effect of return policy leniency on purchase proclivity and return proclivity (based on meta-analysis of 21 papers).

	Number of samples in each analysis (k)	Number of observations in the k samples (N)	Mean uncorrected correlation (r)	Mean weighted corrected correlation (r_t)	Estimated population variance (σ_t^2)	95 percent bootstrapped confidence intervals (CI_{BS})	Effect size	Variance unaccounted for across samples (χ^2)	Fail-safe N for each variable (N_{fsR})
Purchase proclivity	75	21,052	0.212	**0.217****	0.139	0.175 0.261	Small	469.25	22,297
Return proclivity	79	20,831	0.053	**0.080****	0.203	0.021 0.137	Small	1,859.82	569.8

** $p < .01$.

study jk. e_{ij} represents the sampling error of effect size Y_{ij}, which is assumed to be normally distributed with mean of 0 and known variance V_i—which is calculated based on statistics collected from the included study.

$$Y_{ij} = \pi_{0j} + \sum_{p=1}^{P} \pi_{pj} a_{pij} + e_{ij} \quad (1)$$

2. Level 2—Independent Sample Level Model

At level 2, we model the intercept of study j as a function of the retailer characteristics and study control variables that vary across studies $X_{(q=1-8)}$ (*Durable Good*, *Specialty Store*, *Online*, *Elite Journal Publication*, *U.S. Respondents*, *Student Respondents*, *Experimental Setting*). Again, mean centering the variables at level 2 means the intercept, β_{00} is interpreted as the average effect size. The error term r_{0j} is assumed to be normally distributed with mean of 0 and variance τ_π.

$$\pi_{0j} = \beta_{00} + \sum_{q=1}^{Q} \beta_{0q} X_{qj} + r_{0j} \quad (2)$$

Results

Overall Results. The strength of meta-analysis lies in its pooling of estimates of association across studies and analyzing differences in relationship strength according to factors that can distinguish these effects. Table 3 provides an overview of the main effects associated with return policy leniency on purchase indicators and return indicators.

The overall correlation between return leniency and purchase indicators is 0.218 and that between return leniency and return proclivity is 0.080, suggesting that return leniency is associated with both increased purchase proclivity and increased return proclivity. Further, the effect on purchase proclivity is greater than the effect on return proclivity. Consistent with Rosenthal and Rosnow (1991), these effects are categorized as small and are significant. The high N_{FS} for both effects suggests that publication bias is not a problem. Given the heterogeneity present within the dataset ($\chi^2_{pur} = 469.25$, $df = 74$, $p < .01$) ($\chi^2_{ret} = 1859.82$, $df = 78$, $p < .01$), an examination of the moderators of the relationship between return leniency and purchase and return is warranted.

Multivariate Analysis using HLM. The multi-level model as specified in Eqs. (1) and (2) that accounts for multiple effects per study was used to estimate the impact of the influencing variables on purchase proclivity and return proclivity. The meta-regression models of purchase and return proclivites on the influencing variables are provided in Table 4. The reported results are estimated using Restricted Maximum Likelihood techniques, which are more appropriate for estimating the variance parameters due to the small number of level two units (independent samples) in our dataset (Raudenbush and Bryk 2002), and provide the results of a fully conditional model for both purchase and returns. The variables were centered around the grand mean which allows interpretation of the intercept as the adjusted mean effect size for study j.[5]

Table 4
Estimation results—HLM model—fixed effects for each DV.

	Purchase proclivity	Return proclivity
Level 1 Coefficients		
Return Policy Factors		
Time	0.028	**−0.381***
Money	**0.089***	0.021
Effort	**0.122***	−0.156
Scope	0.163	**0.049**+**
Exchange	0.040	**−0.582***
Level 2 Coefficients		
Intercept	**0.243***	**0.111***
Substantive Moderators		
Type of Product	**0.155**+**	0.102
Assortment Focus	0.140	−0.264
Channel Focus	−0.013	−0.457
Methodological Moderators		
Journal Quality	**0.213**+**	−0.019
Research Design	0.085	**−0.726**+**
Study Location	−0.016	0.198
Participant Details	−0.068	**0.195**+**
Variance Component u0	**0.015***	**0.048***
Variance Explained	35.9 percent	41.7 percent
AIC	434.70	752.83
BIC	467.10	783.63

* $p < .05$.
+ $p < .1$.

[5] The simple full conditional model was found to be significantly better (based on AIC/BIC) than a non-conditional model that did not include any of the variables, or a partial conditional model that included only the return policy variables and ignored the moderator variables. Further, within the full conditional model we ran different models by dropping variables such as *Specialty* and *Online*. This did not change the direction or significance of the results. We therefore report only the full conditional model because it captures the effect of all the return policy variables and the moderator variables proposed by prior research.

In examining the dependent variable of purchase proclivity, HLM results reveal that leniency on *money* ($\beta_{pur} = 0.089$, $p < .05$) and *effort* ($\beta_{pur} = 0.122$, $p < .05$) return policy factors lead to significantly higher purchase as compared to leniency on *time* ($\beta_{pur} = 0.028$, $p =$ ns), *scope* ($\beta_{pur} = 0.163$, $p =$ ns) or *exchange* ($\beta_{pur} = 0.040$, $p =$ ns). Further, leniency on durable products ($\beta_{pur} = 0.155$, $p < .05$) showed a higher effect on purchase compared with non-durable products.[6]

An examination of return proclivity as the dependent variable reveals leniency on *time* ($\beta_{ret} = -0.381$, $p < .05$) and *exchange* ($\beta_{ret} = -0.582$, $p < .05$) have a significantly lower effect on returns, while *scope* ($\beta_{ret} = 0.049$, $p < .1$) has a directionally higher effect on returns, as compared to *money* ($\beta_{ret} = 0.021$, $p =$ ns) and *effort* ($\beta_{ret} = -0.156$, $p =$ ns). Among the methodological moderators, experimental studies produce directionally lower returns ($\beta_{ret} = -0.726$, $p < .1$) than field studies, and studies conducted with student samples are directionally higher than those with non-student samples ($\beta_{ret} = 0.195$, $p < .10$).

General Discussion

Retailers often use lenient return policies—possibly due to the belief that leniency inherently increases purchase more than returns—despite the substantial expense associated with product returns. Through meta-analysis, we are able to shed new light on this trade-off. Consistent with Petersen and Kumar (2010), we observe a more pronounced increase in purchase stemming from lenient return policies than for return rates. This suggests that, overall, return policies do in fact benefit retailers—at least in terms of encouraging product purchase.

Our research represents the first attempt to understand the return policy literature in a quantitative, systematic manner using meta-analysis, and to substantiate beliefs that return leniency positively affects both purchase and return decisions. It also challenges the underlying assumption in most previous return policy research—that all return policy factors are similar in their effect on purchase and returns and, therefore, any one or more of the return policy factors can be randomly varied in order to manipulate return policy leniency.

In fact, our consolidation of prior research suggests retailers tend to vary five factors in their return policies to impose restrictions that dissuade returns or offer leniency to encourage purchase. These factors are monetary leniency, time leniency, effort leniency, scope leniency, and exchange leniency. Prior research exploring the effects of lenient return policies yields mixed results, primarily because most work in this area examines only one or two of the various lenient return policy factors. By systematically coding all of the variables for each paper in our meta-analysis database, we are able to provide more clarity on this topic.

Our results imply that careful thought should guide the choice of leniency factors because they have differential effects on purchase and return. To illustrate, imagine two return policies—Policy A (100 percent money back, 30-day deadline) and Policy B (80 percent money back, 60-day deadline)—that vary in terms of money and time. Are both return policies likely to affect purchase and returns equally? While intuition suggests different retailers are sensitive to different aspects of return policy, extant research has mostly assumed that the leniency dimensions affect purchase and return in a similar manner. Our findings show that having both money and time as part of the return policy mean that Policy A will likely affect purchase more, while Policy B will influence returns more. Thus, leniency should be offered on return policy factors depending on the objectives of the retailer.

Specifically, if the retailer's strategy is to stimulate purchase, offering more lenient monetary and/or lower effort return-policies may be more effective. This is in line with theoretical models developed by Shulman, Coughlan, and Savaskan (2010), who examine the effect of restocking fees and find that lower return penalty increases the expected utility of initially making a purchase. We also find that consumers are sensitive to future return restrictions (effort) and incorporate this into their purchase decisions, and less return hassle (e.g., not requiring tags or forms to be filled out) does increase purchase.

However, if retailers wish to curb returns, offering a restrictive return policy in terms of limiting the scope of products eligible for return, in combination with longer deadlines and leniency on exchange, would be more effective. The effect of longer deadlines is in line with findings by Janakiraman and Ordóñez (2012), who find that longer deadlines increase the endowment effect for products and lead to consumers postponing or delaying return decisions. Allowing returns on products purchased during sale periods increases returns. This suggests enforcing restrictions on more price sensitive segments of consumers (who presumably purchase the discounted products) might dissuade return tendency. Leniency on exchange shows a significant and negative effect on product returns, suggesting exchange leniency (i.e., offering cash back in exchange for the returned item) results in fewer returns than restrictive return policies (i.e., offering product exchanges, but no monetary refund). One potential explanation may be that consumers with minor product grievances (enough to trigger exchanges but not returns) are more likely to exchange products when exchange clauses are made salient.

A final implication for retailers, given our finding that return policy leniency increases purchase for durable products markedly more than for non-durable products, is that different return policies should be devised for different categories of products sold in the store, rather than implementing a blanket return policy across all items in the store.

From a theoretical standpoint, our research also suggests that product return decisions are more complex than product purchase decisions because the effect of the different leniency factors is likely to vary not just on the degree of their effect on return, but also on the valence or the direction of their effect on return decisions. Using an HLM model that controls for multiple effects reported in each study, we find that study-level factors

[6] The statistically significant variables in HLM were also found to be significant in univariate analyses—which is not reported in the paper since it essentially replicates the HLM results.

do matter, in that they moderate both the purchase and return decision. Specifically, we find that studies in elite publications are likely to show a higher rate of purchase under lenient return conditions and that studies conducted with student samples are likely to suppress returns. We also find that utilizing an experimental design produces no significant impact with respect to purchase behavior; however, it diminishes return propensity more than field studies.

While our research makes important practical and theoretical contributions, it is fraught with some limitations. First, we acknowledge that return policy research is an evolving field. As such, the emergence of new studies in this area may allow for a meta-analysis of more nuanced constructs (e.g., purchase behavior) rather than broad categories of outcomes (i.e., purchase proclivity), which would potentially lead to greater clarity on the effect of different leniency factors. Second, the studies and the range of variation within each construct that has been examined by prior research limit any meta-analysis, including ours. A third limitation of our research lies in our inability to examine return policy leniency factors in a controlled setting, varying one at a time (in a field study) or together (in a conjoint study), to increase the validity of the results. Further, scant exant research has examined the process by which some of these leniency factors influence purchase, return, and re-purchase decisions. We leave these initiatives to future research.

Our investigation focuses on two dependent variables (i.e., purchase and returns), that could themselves be part of a two-stage process wherein purchase decisions precede return decisions and strong versus weak purchase attitudes subsequently influence return attitudes. Prior research has largely examined these effects separately, with only six of the 31 studies comprising our meta-analytic dataset providing effect sizes for both purchases and returns (i.e., studies in which the same respondents provide responses leading to the measures of the impact of return policy leniency). Because almost two-thirds of the studies in our dataset reported only a purchase or a return decision, we chose to analyze the dependent variables separately. Kalaian and Raudenbush (1996) suggest that if some measure of correlation between the multiple dependent variables is known a priori, then they can be included in the analysis. As the return policy research domain continues to expand and studies emerge that report both purchase and return decisions, we hope future research will be able to examine the correlated nature of these two outcomes.

Further, future research might examine the effect of interactions between the different return policy factors. For example, Shulman, Coughlan, and Savaskan (2010) suggest return penalties (e.g., restocking fees or restricting a monetary leniency factor) are imposed based on the salvage value of the product, which varies by channel member. Thus, understanding the joint effect of exchange leniency and monetary leniency may be important in making these decisions. Along similar lines, Neiman Marcus uses a sliding scale of restocking fees (increasing with the length of ownership) for returns, suggesting that the effect of monetary leniency is contingent on time leniency.

Executive summary

Retailers often use lenient return policies—possibly due to the inherent belief that leniency increases product purchase more than product returns—despite the high processing costs and low salvage values associated with returns. However, previous research exploring the effect of lenient return policies on purchase and subsequent returns is inconclusive. We shed new light on this topic by conducting a meta-analysis in which data is synthesized from original studies examining how return policy leniency affects two important downstream outcomes: (1) purchase proclivity (i.e., purchase attitudes or behaviors) and (2) return proclivity (i.e., consumer intentions or decisions to keep or return products). Our analysis reveals a more pronounced increase in purchase stemming from lenient return policies than for return rates. This suggests that, overall, return policies do in fact benefit retailers—at least in terms of encouraging product purchase.

Additionally, our work challenges the underlying assumption in most prior return policy research—that all forms of return policies affect product purchase and returns in a similar manner. Our consolidation of previous research suggests this is not the case, as retailers tend to impose restrictions that dissuade returns or offer leniency to encourage purchase through the use of five return policy elements. We systematically code the effects of return policy leniency (on product purchase and return) reported in studies included in our meta-analytic database as varying along five leniency dimensions representative of those elements: time leniency (e.g., 60 day vs. 30 day return policy), monetary leniency (e.g., offering 100 percent money back vs. 80 percent money back), effort leniency (e.g., no forms required vs. forms required), scope leniency (e.g., accepting returns on sale items vs. not), and exchange leniency (e.g., cash back vs. store credit). We find these five factors differ in their impact on consumer purchase and return decisions and suggest the mixed results in previous research regarding the effects of return policy leniency are likely due to the fact that most work in this area examines only one or two of the various leniency factors.

Specifically, we find that return policies that offer more monetary and/or effort leniency are likely to increase consumer purchase and, hence, are more likely to drive demand. However, in situations in which the priority is to curb returns, retailers should focus more on other aspects such as return policy deadlines, exchange leniency, and/or scope leniency. Due to the differential influences of the leniency factors, retailers may be better served by creating more complex return policies that vary along multiple dimensions rather than simplistic return policies that vary along only one or two dimensions.

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